Final Evaluation and Evidence Pack

University Student-Support Case Agent

Group M Evening

BSE4104 Emerging Trends in Software Engineering

23 October 2026

1. How to read this pack

This pack is the index for the evaluation evidence. The detailed numbers stay in the files they were first recorded in. Week 7 totals are not rewritten after the Week 8 fixes. A reader who wants the original score uses week7_evaluation.md. A reader who wants the repair uses the failure catalogue and week8_retest.json.

2. Evidence map

Foundation model

gemini-3.6-flash in src/tools/orchestrate.py and src/llm_client.py. Rationale in the model selection note.

Prompts

prompts/v1.0_baseline.txt through prompts/v1.3_tools.txt, plus the prompt specification PDF.

RAG

Corpus register, data/chunks/chunks.json, src/rag/retrieve.py, docs/evaluation/rag_eval.md. Result: 11/15 (73%).

Tools

retrieve_documents and create_support_case. Tool catalogue PDF and the MCP-style specification.

Bounded agent

src/tools/orchestrate.py. Stops S1-S8. Traces evidence/traces/T1, T2, and T3.

State and memory

src/agent/state.py, Case History Recall, state-model PDF, memory note.

Evaluation

36 scenarios, 29/36 (80.5%). week7_evaluation.md and the scenario workbook.

Observability

src/observability/turn_log.py. Samples O1-O4. Observability PDF.

Guardrails

Sense stops, validation, agreement flag, tool allow-list. Guardrail evidence PDF.

Week 8 retest

16/16 deterministic checks. The code check is 5/5. Live run: B07 and N03 pass, B06 is refused by the model, and I02 still misses.

AI Engineering Log

Appendix C of the final engineering report. gemini-embedding-001, gemini-3.6-flash, and Cursor. No secrets or student records were sent.

3. Week 8 retest record

Command, from src/, with no API key: python run_deterministic_checks.py. Result: 16 PASS, 0 FAIL. The checks that had been failing on the Week 7 code were C07 (fee decision under a non-fee category), C08 (placeholder summary), and R04 (raw 503 text). R05 still expects the case id to be named when a draft was saved before the model error. That check passed.

Command: python week8_retest.py. Result: 5 PASS, 0 FAIL. B06, the Code of Conduct question, is not blocked. B07, the admission-letter documents question, is not blocked. "I need help with admissions for next year" is still blocked. "Please change my grade" is still blocked. "I'm a second-year student and need the Wi-Fi password." is still out of scope.

4. Live retest on 24 October 2026

Nineteen live chats were run from docs/evaluation/week8_runner_input.xlsx. Eighteen finished. The replies are in docs/evaluation/week8_live_results.csv. B07 passes with a cited list of registration documents. N03 stops at S4. I02 still returns no_match and stops at S3. B06 reaches the model, which refuses the Code of Conduct question without searching. M01 did not finish: Gemini returned 429 RESOURCE_EXHAUSTED. The Week 7 score of 29/36 is not rewritten. The results file used for this run has 35 rows.

During Week 7 observability, gemini-3.6-flash returned 503 under load. O4 stores that 503. The student-facing path for the same class of error is now the short busy message, which R04 checks without calling the network. The live runner treats that busy sentence as an infrastructure error and retries it. M01 still exhausted the free-tier quota after those retries.

I02 and Week 3 Q5 remain retrieval misses at the 0.70 cutoff. No new similarity number is claimed for them in Week 8, because the index was not rebuilt and the questions were not re-embedded.

5. Reproduce

- git checkout week8-release, at commit 8139dc3. Tag v1.5.0 is the earlier hardened release and does not include this live retest.
- pip install -r requirements.txt
- Copy .env.example to .env and set GEMINI_API_KEY for any live run.
- From src/rag/: download_sources.py, extract_text.py, chunk_text.py, embed_chunks.py.
- From the repo root: python -m streamlit run src/app.py
- From src/: python run_deterministic_checks.py
- From src/: python week8_retest.py
- Live chats, uses quota: python run_scenarios.py ../docs/evaluation/week8_runner_input.xlsx --repeat 1